6.2 Bisectors of Triangles

Geometry 6.2 · Student sheet

Name _____ Period ____ Date _____

I can…

- use the Circumcenter Theorem to find distances to the vertices.
- find the circumcenter of a triangle in the coordinate plane.
- use the Incenter Theorem to find distances to the sides.

Vocabulary · fill in the missing word

Concurrent — three or more lines, rays, or segments that _____ in the same point

Point of concurrency — the point where concurrent lines, rays, or segments _____

Circumcenter — the point of concurrency of the _____ bisectors of a triangle

Incenter — the point of _____ of the angle bisectors of a triangle

Part 1 · Discovery: fold the bisectors

Fold the perpendicular bisectors (vertex onto vertex), then the angle bisectors (fold each angle in half).

Creases	Meet at one point?	Distances measured	What is equal?	Inside or outside?
Perpendicular bisectors (acute)				
Perpendicular bisectors (obtuse)				
Angle bisectors				

Part 2 · Notes and examples (fill in each note, then work the example)

Circumcenter Theorem: the circumcenter of a triangle is equidistant from the _____ of the triangle.

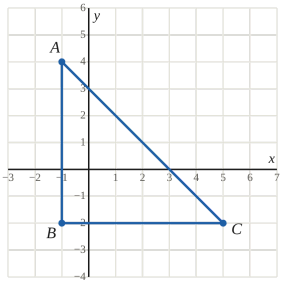
Example 1

Point P is the circumcenter of $\triangle ABC$. $PA = 3x + 2$ and $PB = 5x - 8$. Find x and PC.

Where the circumcenter is: inside an acute triangle, at the midpoint of the hypotenuse of a right triangle, and _____ an obtuse triangle.

Example 2

Find the circumcenter of the triangle with vertices A(−1, 4), B(−1, −2), and C(5, −2).



Incenter Theorem: the incenter of a triangle is equidistant from the _____ of the triangle, and it is always inside.

Example 3

Point N is the incenter of $\triangle ABC$. D, E, and F are where the perpendicular segments from N meet the sides. $ND = 2x - 1$ and $NE = x + 4$. Find x and NF.

Example 4

The angle bisectors of $\triangle XYZ$ meet at P. Point K is on $\overline{XY}$ with $\overline{PK} \perp \overline{XY}$. $PX = 13$ and $XK = 12$. Find the distance from P to $\overline{YZ}$.

Which point? The same distance from three points: the circumcenter. The same distance from three lines or sides: the _____.

Example 5

Which point of concurrency should you use?

a. A fire station the same distance from three towns

b. A fountain the same distance from the three straight paths around a triangular park

Part 3 · Partner practice

Partner 1

Point P is the circumcenter of $\triangle ABC$. $PA = 4x - 3$ and $PC = 2x + 9$. Find PB.

Partner 2

The angle bisectors of $\triangle ABC$ meet at P. Point K is on $\overline{AB}$ with $\overline{PK} \perp \overline{AB}$. $PA = 10$ and $AK = 8$. Find the distance from P to $\overline{BC}$.